INDIAN INSTITUTE OF INFORMATION TECHNOLOGY ALLAHABAD

Department of Information Technology

IMAGE & VIDEO PROCESSING

Software Requirements Specification
& Design Report

for

Multi-Feature Fusion Based Cross-Camera Person Tracking System

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Base Research Paper

Cross-Camera Tracking Model and Method Based on Multi-Feature Fusion

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Prepared By

S.No.	Name	Roll Number
1	Vardaan Gupta	IIT2024046
2	Parth Khatri	IIT2024002
3	Hemant Singh	IIT2024047
4	Aryan Saini	IIT2024008
5	Anshdeep Singh	IIT2024011
6	Udaynoor Singh	IIT2024006
7	Aditya Sirsalkar	IIT2024009

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) and Design Report is to define the requirements and high-level design of the proposed **Multi-Feature Fusion Based Cross-Camera Person Tracking System**.

The proposed system aims to detect, track, and maintain the identity of a person across multiple camera views. Unlike conventional single-camera tracking, the proposed system attempts to associate observations of the same individual when that individual moves from one camera view to another.

1.2 Objectives

The primary objective of this project is to design and develop a **Multi-Feature Fusion Based Cross-Camera Person Tracking System** for maintaining consistent person identities across multiple camera views.

The specific objectives of the proposed project are as follows:

1. **Multi-Camera Person Detection:** To detect persons from video streams captured by multiple cameras.
2. **Single-Camera Person Tracking:** To track detected persons within an individual camera view and maintain consistent local identities across consecutive video frames.
3. **Feature Extraction:** To extract discriminative appearance features along with motion and trajectory-related information for each tracked person.
4. **Multi-Feature Fusion:** To combine multiple complementary features to obtain a more robust representation of a person.
5. **Cross-Camera Association:** To identify and associate observations of the same person appearing in different camera views.
6. **Global Identity Management:** To assign and maintain a consistent global identity for a person across multiple cameras.
7. **Performance Improvement:** To reduce identity switches and improve the reliability of cross-camera tracking under challenges such as viewpoint variations, occlusion, and appearance changes.
8. **Visualization and Analysis:** To provide visualization of tracking results and support the analysis of detection, tracking, and cross-camera association performance.

1.3 Scope

The system will process multiple video streams and perform person detection, tracking, feature extraction, feature fusion, and cross-camera association.

The project scope includes:

- Processing multiple surveillance video streams.

- Detecting persons in individual video frames.
- Tracking persons within individual cameras.
- Assigning local identities to tracked persons.
- Extracting appearance-related features.
- Extracting trajectory and motion information.
- Combining multiple features.
- Identifying the same person across different cameras.
- Maintaining consistent global identities.
- Displaying and storing tracking results.

1.4 Definitions, Acronyms and Abbreviations

Term	Definition
SRS	Software Requirements Specification
IVP	Image & Video Processing
MTMCT	Multi-Target Multi-Camera Tracking
Re-ID	Person Re-Identification
ROI	Region of Interest
IoU	Intersection over Union
IDF1	Identity F1 score used for tracking evaluation
Local ID	Identity assigned to a person within one camera
Global ID	Identity maintained across multiple cameras
Feature Fusion	Combining multiple features to improve matching

1.5 Base Research Paper

The proposed project is primarily based on the following research paper:

> P. Zhang, S. Wang, W. Zhang, W. Lei, X. Zhao, Q. Jing, and M. Liu, "Cross-Camera Tracking Model and Method Based on Multi-Feature Fusion," *Symmetry*, vol. 15, no. 12, Article 2145, 2023.

DOI: https://doi.org/10.3390/sym15122145

The paper provides the conceptual foundation for combining multiple features to improve the association of persons across different camera views.

1.6 Document Overview

This document describes the project scope, objectives, base research paper, overall system characteristics, functional requirements, non-functional requirements, data requirements, architecture, block diagram, system workflow, flowchart, and verification requirements.

2. Overall Description

2.1 Product Perspective

The proposed system is a computer vision-based application designed for cross-camera person tracking. It follows a modular architecture consisting of:

1. Video Input Module
2. Person Detection Module
3. Single-Camera Tracking Module
4. Feature Extraction Module
5. Trajectory and Motion Analysis Module
6. Multi-Feature Fusion Module
7. Cross-Camera Association Module
8. Global Identity Management Module
9. Visualization Module

Each camera stream is initially processed independently. The resulting person observations are then compared and associated across different cameras.

2.2 Product Functions

The major functions of the proposed system are:

- Receive multiple video inputs.
- Detect persons in video frames.
- Track persons across consecutive frames.
- Assign and maintain local identities.
- Generate person trajectories.
- Extract appearance features.
- Extract motion and temporal information.
- Combine multiple features.
- Compare persons across different cameras.
- Assign global identities.
- Visualize tracking information.

2.3 User Classes and Characteristics

2.3.1 Research User

The research user provides datasets, configures experimental parameters, runs the system, and analyzes results.

2.3.2 System Operator

The system operator manages camera inputs and initiates video processing.

2.3.3 Viewer

The viewer observes tracking results generated by the system.

2.4 Operating Environment

The proposed system is expected to operate in:

- Operating System: Linux or Windows.
- Programming Language: Python.
- Computer Vision Library: OpenCV.
- Deep Learning Framework: PyTorch.
- Input: Recorded videos or live camera streams.
- Hardware: GPU-enabled system recommended.

2.5 Design and Implementation Constraints

- Performance depends on video quality.
- Severe occlusion may affect tracking accuracy.
- Large viewpoint changes may affect person matching.
- Real-time performance depends on computational resources.
- Processing multiple cameras requires additional computation.

2.6 Assumptions and Dependencies

The system assumes that:

- Valid video streams are available.
- Persons remain visible for a sufficient number of frames.
- Video quality is sufficient for person detection.
- Required detection and re-identification models are available.
- Adequate computational resources are available.

3. External Interface Requirements

3.1 User Interface

The system interface shall provide functionality to:

- Select video inputs.
- Start and stop processing.
- View processed camera streams.
- View person bounding boxes.
- View local and global identities.
- Observe tracking information.

3.2 Hardware Interfaces

The system may interface with:

- CCTV cameras.
- USB cameras.
- IP cameras.
- GPU-enabled computing systems.

3.3 Software Interfaces

The system may use:

- OpenCV for video processing.
- PyTorch for deep learning operations.
- Person detection models.
- Person re-identification models.
- Multi-object tracking algorithms.

3.4 Communication Interfaces

The system may receive video through:

- Local video files.
- Network video streams.
- Real-Time Streaming Protocol (RTSP).

4. Functional Requirements

4.1 Video Input Management

ID	Priority	Requirement
FR-01	High	The system shall accept input from multiple video streams or recorded videos.
FR-02	High	The system shall support commonly used video formats such as MP4 and AVI.
FR-03	High	The system shall process individual camera streams before cross-camera association.

4.2 Person Detection

ID	Priority	Requirement
FR-04	High	The system shall detect persons in each processed video frame.
FR-05	High	The system shall generate bounding boxes for detected persons.
FR-06	Medium	The system shall associate a confidence value with each detection.

4.3 Single-Camera Tracking

ID	Priority	Requirement
FR-07	High	The system shall assign a unique local identity to each tracked person within an individual camera stream.
FR-08	High	The system shall maintain the identity of a tracked person across consecutive frames.
FR-09	High	The system shall generate and maintain the trajectory of each tracked person.

4.4 Feature Extraction

ID	Priority	Requirement
FR-10	High	The system shall extract appearance features for each tracked person.
FR-11	High	The system shall extract trajectory-related and motion-related information.
FR-12	Medium	The system shall record temporal information related to a person's appearance in a camera.

4.5 Multi-Feature Fusion

ID	Priority	Requirement
FR-13	High	The system shall combine appearance and motion-related features to generate a fused representation.
FR-14	Medium	The system should support configurable feature weights for experimental analysis.

A conceptual representation of feature fusion is:

$$F_{\text{fusion}} = w_a F_{\text{appearance}} + w_m F_{\text{motion}}$$

where w_a and w_m represent the respective feature weights.

4.6 Cross-Camera Association

ID	Priority	Requirement
FR-15	High	The system shall identify potential matches between persons observed in different camera streams.
FR-16	High	The system shall calculate similarity between candidate observations using fused features.
FR-17	High	The system shall associate observations determined to represent the same person.
FR-18	Medium	The system should support spatial or coordinate transformation when camera calibration information is available.

4.7 Global Identity Management

ID	Priority	Requirement
FR-19	High	The system shall assign a global identity to successfully associated person observations.
FR-20	High	The system shall maintain a consistent global identity across multiple cameras.
FR-21	Medium	The system shall maintain a history of camera appearances associated with each global identity.

4.8 Visualization and Reporting

ID	Priority	Requirement
FR-22	High	The system shall display detected persons using bounding boxes and identifiers.
FR-23	Medium	The system shall support simultaneous visualization of multiple camera streams.
FR-24	Medium	The system should generate tracking statistics and association results.

5. Non-Functional Requirements

5.1 Performance Requirements

- The system should process video frames with minimum possible latency.
- The system should utilize GPU acceleration when supported hardware is available.
- The system should support processing multiple camera streams.

5.2 Reliability Requirements

- The system should minimize unnecessary identity switches.
- The system shall handle invalid video inputs without unexpected termination.

5.3 Scalability Requirements

The architecture should support the addition of new camera streams with minimal modification.

5.4 Usability Requirements

The user interface should allow users to perform basic operations with minimal technical complexity.

5.5 Maintainability Requirements

The system shall use a modular architecture to allow independent modification of major components.

5.6 Portability Requirements

The system should be deployable on common Linux and Windows environments.

6. Data Requirements

6.1 Input Data

The system may receive:

- Recorded surveillance videos.
- Multiple camera streams.
- Live camera feeds.
- Camera configuration information.

6.2 Generated Data

The system shall generate:

- Bounding box coordinates.
- Local identities.
- Global identities.
- Appearance feature vectors.
- Motion information.
- Person trajectories.
- Similarity scores.
- Cross-camera association results.

6.3 Data Storage

The system may store tracking information and experimental results in structured files or a database.

7. System Workflow

The high-level workflow of the proposed system is:

Multiple Video Streams

↓

Person Detection

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Single-Camera Tracking

↓

Feature and Trajectory Extraction

↓

Multi-Feature Fusion

↓

Cross-Camera Association

↓

Global Identity Assignment

↓

Visualization and Analysis

8. System Architecture and Design

This section presents the system architecture, detailed processing blocks, and operational flow of the proposed project.

8.1 System Architecture Diagram

The proposed system follows a layered and modular architecture.

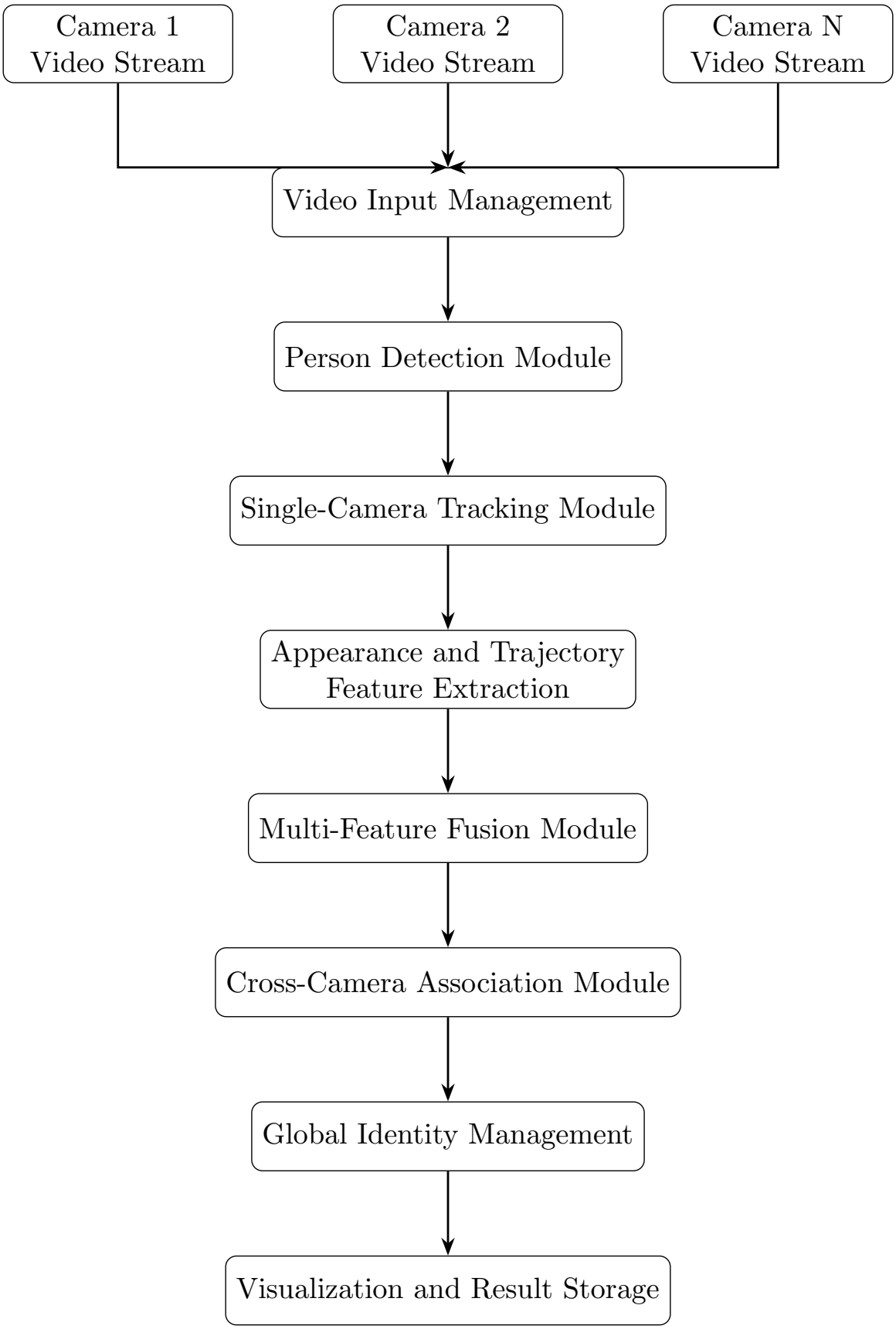


Figure 1: High-Level System Architecture

8.2 Processing Block Diagram

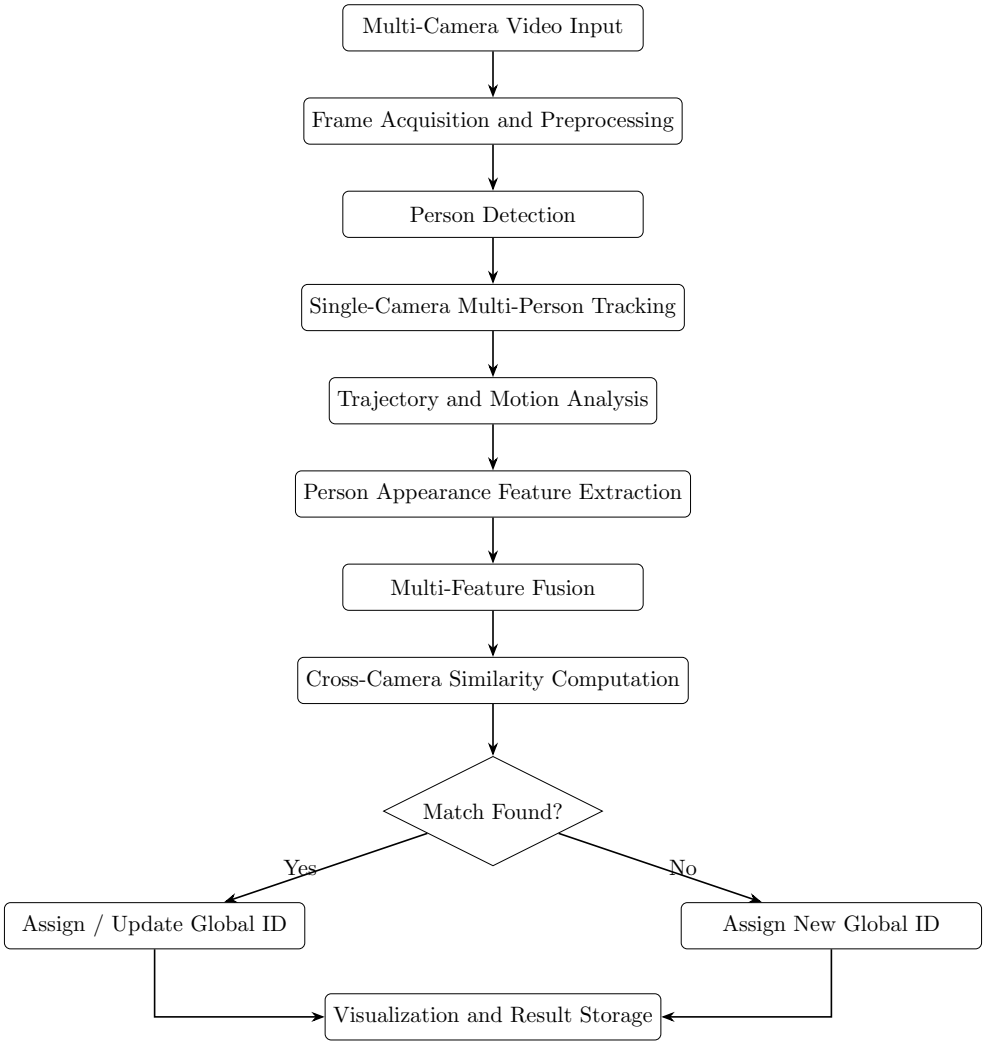


Figure 2: Detailed Processing Block Diagram

8.3 System Flowchart

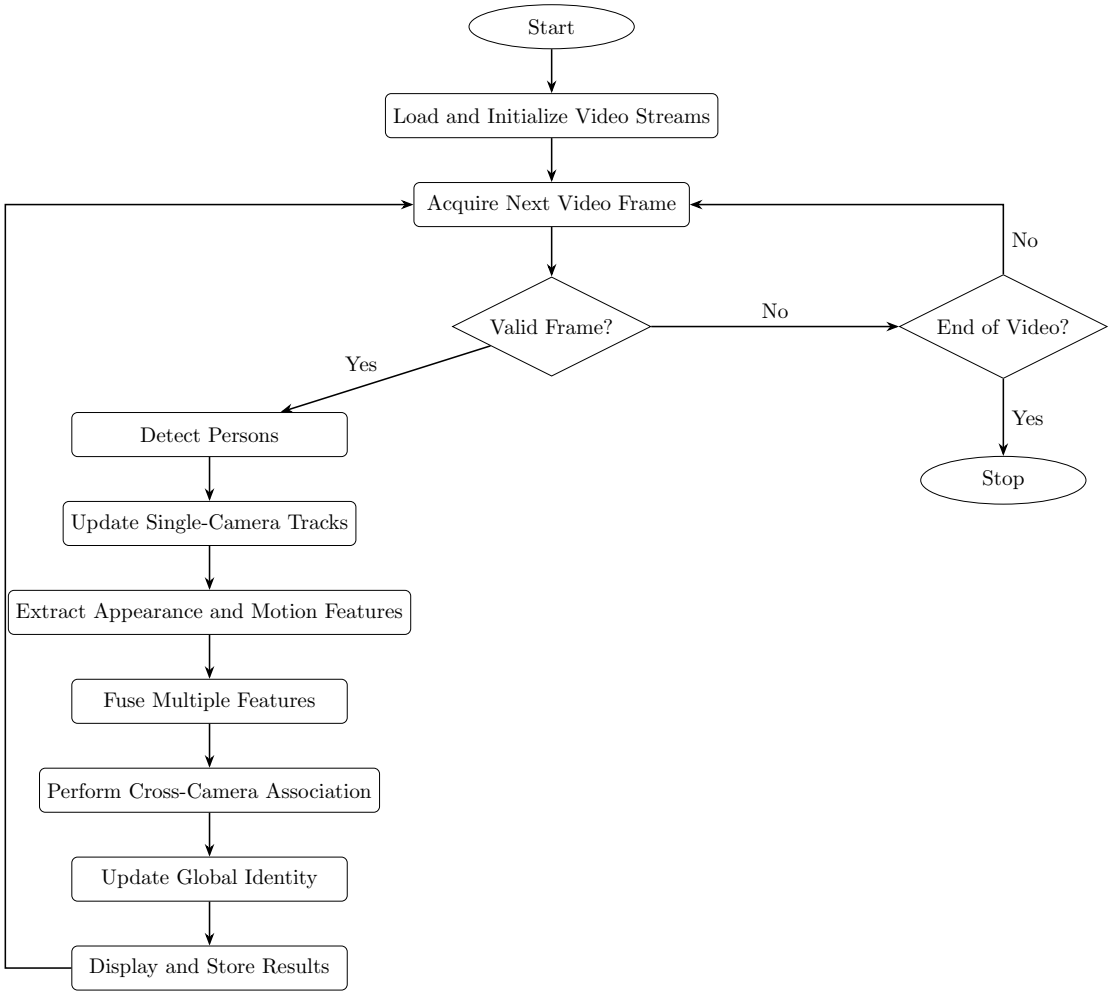


Figure 3: Operational Flowchart of the Proposed System

8.4 Module Interaction

The interaction between the major modules is summarized below:

1. The video input module receives multiple camera streams.
2. Frames are sent to the person detection module.
3. Detected persons are passed to the single-camera tracking module.
4. Local identities and trajectories are generated.
5. Appearance and motion features are extracted.
6. The extracted features are fused.
7. Candidate observations are compared across cameras.
8. Global identities are assigned or updated.
9. Final results are visualized and stored.

9. Verification and Validation Requirements

9.1 Functional Testing

Each functional requirement shall be tested to verify that the corresponding functionality is correctly implemented.

9.2 Detection Evaluation

Person detection performance may be evaluated using standard detection metrics.

9.3 Tracking Evaluation

Tracking performance may be evaluated using:

- IDF1 Score.
- Identity Precision.
- Identity Recall.
- Number of Identity Switches.

9.4 Cross-Camera Association Evaluation

The system shall be evaluated based on its ability to correctly identify and associate the same person across different camera views.

9.5 Performance Testing

The system shall be tested using different video inputs and varying numbers of camera streams.

10. Future Enhancements

Possible future enhancements include:

- Real-time deployment using live surveillance cameras.
- Support for larger camera networks.
- Adaptive feature weighting.
- Improved deep learning-based person re-identification.
- Automated camera topology learning.
- Edge-device deployment.
- Advanced real-time analytics dashboards.